Chapter 7 – Results and Discussion

7.1 Introduction

This chapter presents the evaluation and validation of the Intelligent Task Allocation System (ITAS) in terms of functionality, performance, usability, and alignment with the project objectives. The evaluation process focuses on assessing the effectiveness of the machine learning-based task allocation mechanism, the reliability of the matching engine, and the overall user experience provided to project managers and employees.

The chapter also discusses the data collection and preprocessing methodologies used during the development of the skill classification model. In addition, both functional and non-functional testing results are presented alongside feedback collected during the UI/UX evaluation phase to determine the practicality and usability of the system in real-world project management scenarios.

7.2 Data Collection and Research

The core functionality of ITAS depends heavily on accurate resume parsing, skill extraction, and category prediction. To support these capabilities, the system was trained using publicly available resume datasets obtained from Kaggle, specifically the datasets “snehaanbhawal/resume-dataset” and “gauravduttakiit/resume-dataset”. These datasets provided diverse, real-world CV samples from multiple technical domains and professional categories.

Before training the classification model, the collected data underwent several preprocessing stages to improve data quality and consistency. Raw resume text was cleaned using a custom text normalization function designed to remove formatting inconsistencies, unnecessary symbols, and irrelevant noise from the documents.

To improve feature representation and capture both semantic meaning and technical terminology, the system implemented a dual-feature extraction approach using TF-IDF (Term Frequency–Inverse Document Frequency). The first vectorizer operated at the word level using 1–2 n-grams with a maximum of 40,000 features, while the second vectorizer operated at the character level using 3–5 n-grams. Both vectorizers were combined using a FeatureUnion structure to improve the model’s ability to recognize technical skills and domain-specific patterns.

The processed dataset was divided into training and testing subsets using an 80/20 train-test split while maintaining category stratification to preserve dataset balance. A CalibratedClassifierCV model built on top of LinearSVC was then trained and evaluated on unseen data to measure classification reliability and prediction accuracy.

7.3 System Testing and User Feedback

7.3.1 Functional Testing

Functional testing was conducted to verify that all major components of ITAS operated according to the system requirements and design specifications. The machine learning-based classification model was evaluated during training using standard classification metrics including accuracy, precision, recall, and Macro F1-score to ensure reliable text categorization performance.

The MatchingEngine also underwent extensive validation to verify fairness and consistency in task allocation. The engine successfully calculated dynamic suitability scores ranging from 0–100 based on weighted parameters including skill matching, skill coverage, employee experience, workload level, and historical performance ratings.

One of the most significant functional features of the system is the fallback allocation mechanism implemented within the find_best_matches function. In situations where strict SQL-based filtering fails to identify suitable candidates due to niche requirements or incomplete data, the system automatically switches to workload-based filtering instead of returning empty results. This guarantees that project managers always receive candidate recommendations and prevents workflow interruptions.

Additionally, the engine successfully tokenized required skills and calculated overlap ratios between employee skills and task requirements. A minimum overlap threshold of 34% was enforced to ensure that partial matches remained possible while still penalizing weak matches appropriately.

7.3.2 Non-Functional Testing

Non-functional testing focused primarily on performance, scalability, and architectural efficiency. To optimize performance when handling larger employee datasets, ITAS utilized Django ORM filtering and database-level annotations rather than loading all employee records directly into application memory.

By applying relational filters and workload constraints at the database level, the system significantly reduced the candidate pool before executing computationally expensive matching operations. This optimization improved scalability and reduced unnecessary processing overhead during task allocation.

The system also demonstrated architectural flexibility through dynamic weighting logic. Depending on task priority (High, Medium, or Low), the MatchingEngine automatically adjusted the mathematical weights assigned to different employee attributes. For example, high-priority tasks emphasized technical skill matching, while low-priority tasks prioritized workload availability to reduce employee burnout and maintain balanced resource distribution.

7.3.3 User Feedback and UI/UX Evaluation

During the testing phase, the ITAS interface underwent UI/UX evaluation sessions involving beta testers acting primarily in the Project Manager role. Feedback indicated that users found the AI-assisted allocation process efficient, practical, and significantly less biased than manual task assignment approaches.

However, testers also identified several usability improvements related to dashboard readability and score interpretation. In particular, users requested clearer explanations regarding suitability score differences and more effective visualization of employee workload capacity.

Based on this feedback, several non-disruptive UI refinements were implemented, including improvements to typography, visual hierarchy, and dashboard organization. These enhancements increased data readability, reduced user confusion, and improved the overall task assignment workflow.

Overall, the evaluation process demonstrated that ITAS successfully met its primary objectives by providing accurate task allocation, scalable performance, and a user-friendly management interface suitable for collaborative software development environments.